The three-equation melt condition at an ice–water interface

and its form in `solver2d`

1 Geometry and notation

The ice occupies $y < 0$ and the water $0 < y < L_y$; the ice–water interface is the wall $y = 0$ (in `solver2d`: `bc_bot = 'noslip'`, `bc_top = 'freeslip'`, gravity along $+y$). The unit normal $\mathbf{n} = \mathbf{e}_y$ points from the interface into the water. The subscript b denotes values at the interface ($y = 0$). The melt rate m is the speed at which the interface recedes into the ice (in $\mathrm{m\,s^{-1}}$ of ice), positive for melting.

symbol	meaning	typical value
$T,\ S$	water temperature, salinity	
$T_b,\ S_b$	temperature and salinity at the interface	unknown
m	melt rate (ice thickness per unit time)	unknown
p_b	pressure at the interface	
λ_1	liquidus slope	$\approx -5.7 \times 10^{-2}\ ^\circ\mathrm{C\,psu^{-1}}$
λ_2	liquidus offset	$\approx 8.3 \times 10^{-2}\ ^\circ\mathrm{C}$
λ_3	pressure coefficient of the freezing point	$\approx -7.5 \times 10^{-4}\ ^\circ\mathrm{C\,dbar^{-1}}$
L_f	latent heat of fusion	$3.34 \times 10^5\ \mathrm{J\,kg^{-1}}$
$c_w,\ c_i$	specific heat of water, ice	$\approx 3974,\ 2009\ \mathrm{J\,kg^{-1}K^{-1}}$
$\rho_w,\ \rho_i$	density of water, ice	$\approx 1028,\ 917\ \mathrm{kg\,m^{-3}}$
$\kappa_T,\ \kappa_S$	molecular diffusivities of heat and salt in water	$\approx 1.4 \times 10^{-7},\ \approx 7 \times 10^{-10}\ \mathrm{m^2 s^{-1}}$
κ_i	thermal diffusivity of ice	$\approx 1.1 \times 10^{-6}\ \mathrm{m^2 s^{-1}}$
S_i	salinity of the ice	≈ 0

The liquidus constants differ slightly between sources; the values above are representative of those used in ice–ocean models (e.g. Holland & Jenkins 1999). Check them, and the other material properties, against the source you use.

2 The three equations

1. Freezing point (liquidus). The interface is at the local freezing temperature:

$$T_b = \lambda_1 S_b + \lambda_2 + \lambda_3\, p_b. \tag{1}$$

2. Heat balance. The heat conducted from the water to the interface melts ice and is partly conducted into the ice:

$$\rho_w c_w \kappa_T \left.\frac{\partial T}{\partial y}\right|_b = \rho_i L_f\, m \ +\ \rho_i c_i \kappa_i \left.\frac{\partial T_i}{\partial y}\right|_b . \tag{2}$$

The last term (conduction into the ice, T_i the ice temperature) is often neglected, or modelled as $\rho_i c_i \kappa_i (T_b - T_s)/h_i$ with T_s the temperature at the far side of an ice layer of thickness h_i.

3. Salt balance. The ice holds (almost) no salt. The salt that the meltwater dilutes at the interface must be supplied by diffusion in the water:

$$\rho_w \kappa_S \left.\frac{\partial S}{\partial y}\right|_b = \rho_i \, m \, (S_b - S_i) \;\approx\; \rho_i \, m \, S_b. \tag{3}$$

Equations (1)–(3) determine the three unknowns T_b, S_b and m. For melting ($m > 0$) the interface is fresher than the water ($\partial S/\partial y|_b > 0$) and warmer than the freezing point of the far-field water. Since $\kappa_T/\kappa_S \approx 10^2$, salt diffusion limits the melt rate together with the heat supply.

Signs. With $\mathbf{n} = \mathbf{e}_y$ pointing into the water, the conductive fluxes *towards* the interface are $\rho_w c_w \kappa_T \, \partial T/\partial y|_b$ and $\rho_w \kappa_S \, \partial S/\partial y|_b$. This holds for the set-up above: ice at $y = 0$, water at $y > 0$, gravity along $+y$ (the ice is physically on top). If instead the ice is at $y = L_y$ (water at $y < L_y$, normal $-\mathbf{e}_y$), replace $\partial/\partial y$ by $-\partial/\partial y$.

Moving interface. The interface recedes into the ice at speed m. Mass conservation across the interface, $\rho_i \, m = \rho_w \, (m + v_b)$ in the frame of the interface, gives the water velocity at the interface (in the frame of the ice)

$$v_b = \left(\frac{\rho_i}{\rho_w} - 1\right) m, \tag{4}$$

which is negative: since ice is less dense than water, melting contracts the volume and the water moves towards the ice. Both are many orders of magnitude slower than the flow, so most simulations keep the interface fixed and impermeable (quasi-steady approximation), as assumed below.

3 Special cases

Stefan condition (fresh water). Without salt, (3) drops out and (1) becomes $T_b = T_m(p_b)$ ($0\,^\circ\mathrm{C}$ at atmospheric pressure):

$$T_b = T_m, \qquad \rho_i L_f \, m = \rho_w c_w \kappa_T \left.\frac{\partial T}{\partial y}\right|_b - \rho_i c_i \kappa_i \left.\frac{\partial T_i}{\partial y}\right|_b. \tag{5}$$

Transfer-coefficient form (unresolved boundary layers). Ocean models do not resolve the diffusive sublayers and write the fluxes with transfer coefficients γ_T, γ_S and the friction velocity u_*:

$$\kappa_T \left.\frac{\partial T}{\partial y}\right|_b \to \gamma_T u_* (T_w - T_b), \qquad \kappa_S \left.\frac{\partial S}{\partial y}\right|_b \to \gamma_S u_* (S_w - S_b), \tag{6}$$

with T_w, S_w the values in the water outside the sublayer. Neglecting conduction into the ice and with $S_i = 0$, eliminating m and T_b gives a quadratic for S_b:

$$a \, S_b^2 + b \, S_b + c = 0, \qquad \begin{aligned} a &= c_w \gamma_T \lambda_1, \\ b &= -\left[c_w \gamma_T (T_w - \lambda_2 - \lambda_3 p_b) + L_f \gamma_S\right], \\ c &= L_f \gamma_S \, S_w, \end{aligned} \tag{7}$$

Since $\lambda_1 < 0$, $a < 0$ while $c > 0$, so the two roots have opposite signs and the physical root is the positive one (it satisfies $0 < S_b \le S_w$ for melting). Then T_b follows from (1) and $m = (\rho_w/\rho_i) \, \gamma_S u_* (S_w - S_b)/S_b$.

In a resolved simulation the gradients are computed directly at the wall and no transfer coefficients are needed. If the wall gradient is linearised as $\partial T/\partial y|_b \approx (T_1 - T_b)/\delta_T$ (and similarly for S), the same quadratic (7) follows with $\gamma_T u_* \to \kappa_T/\delta_T$, $\gamma_S u_* \to \kappa_S/\delta_S$, $T_w \to T_1$ and $S_w \to S_1$.

4 Non-dimensional form in `solver2d`

`solver2d` uses a length unit ℓ, a velocity unit U, and $Re = U\ell/\nu$. The diffusivities are $\kappa_T = U\ell/(Re\,Pr)$ and $\kappa_S = U\ell/(Re\,Sc)$. Write the scalars as

$$T = T_{\mathrm{ref}} + \Delta T\,\theta, \qquad S = S_{\mathrm{ref}} + \Delta S\,s, \tag{8}$$

(θ and s are scalars 1 and 2 of the code) and the melt rate as a water velocity in units of U:

$$m^* = \frac{\rho_i\, m}{\rho_w\, U}. \tag{9}$$

With y in units of ℓ, equations (1)–(3) become

$$\theta_b = \Lambda\, s_b + \theta_f, \qquad \Lambda = \frac{\lambda_1 \Delta S}{\Delta T}, \quad \theta_f = \frac{\lambda_1 S_{\mathrm{ref}} + \lambda_2 + \lambda_3 p_b - T_{\mathrm{ref}}}{\Delta T}, \tag{10}$$

$$m^* = St\left(\frac{1}{Re\,Pr}\left.\frac{\partial\theta}{\partial y}\right|_b - q_i\right), \quad St = \frac{c_w \Delta T}{L_f}, \tag{11}$$

$$\frac{1}{Re\,Sc}\left.\frac{\partial s}{\partial y}\right|_b = m^*\,(s_b + \sigma), \qquad \sigma = \frac{S_{\mathrm{ref}} - S_i}{\Delta S}, \tag{12}$$

with the Stefan number St and the non-dimensional conduction into the ice

$$q_i = \frac{\rho_i c_i \kappa_i}{\rho_w c_w U \Delta T}\left.\frac{\partial T_i}{\partial y}\right|_b \qquad (q_i = 0 \text{ if conduction into the ice is neglected}). \tag{13}$$

Equation (12) uses $S_b - S_i = \Delta S\,(s_b + \sigma)$. If S_{ref} is chosen as the ice salinity ($S_{\mathrm{ref}} = S_i$, e.g. 0), then $\sigma = 0$.

The new non-dimensional parameters are St, Λ, θ_f, σ (and q_i). In `solver2d` the buoyancy enters through the Richardson numbers `ri(1)`, `ri(2)` of θ and s, and the diffusivities through `pr(1)` $= Pr$ and `pr(2)` $= Sc$.

5 Use as a boundary condition (not yet implemented)

The conditions hold at every point x of the interface and are nonlinear ($m^* s_b$) and coupled (θ and s). A practical implementation in `solver2d`:

1. At each Runge–Kutta substep, evaluate $\partial\theta/\partial y|_b$ and $\partial s/\partial y|_b$ in physical space (all x), from the current fields.
2. At each x, solve (10)–(12) for θ_b, s_b, m^*, e.g. with the linearised wall gradients of Sec. 3, which gives the quadratic (7) at every point.
3. Transform $\theta_b(x)$ and $s_b(x)$ to Fourier space and impose them as Dirichlet values on all Fourier modes in the (batched) scalar solve. The code currently imposes wall values on the mean mode only, so this needs per-mode wall values.
4. Lag the condition by one substep, or iterate it within the substep; the interface values change slowly compared with the flow.
5. Output the melt rate $m^*(x,t)$ and its average.

References

- H. H. Hellmer and D. J. Olbers, A two-dimensional model for the thermohaline circulation under an ice shelf, *Antarctic Science* 1 (1989) 325–336.
- D. M. Holland and A. Jenkins, Modeling thermodynamic ice–ocean interactions at the base of an ice shelf, *J. Phys. Oceanogr.* 29 (1999) 1787–1800.